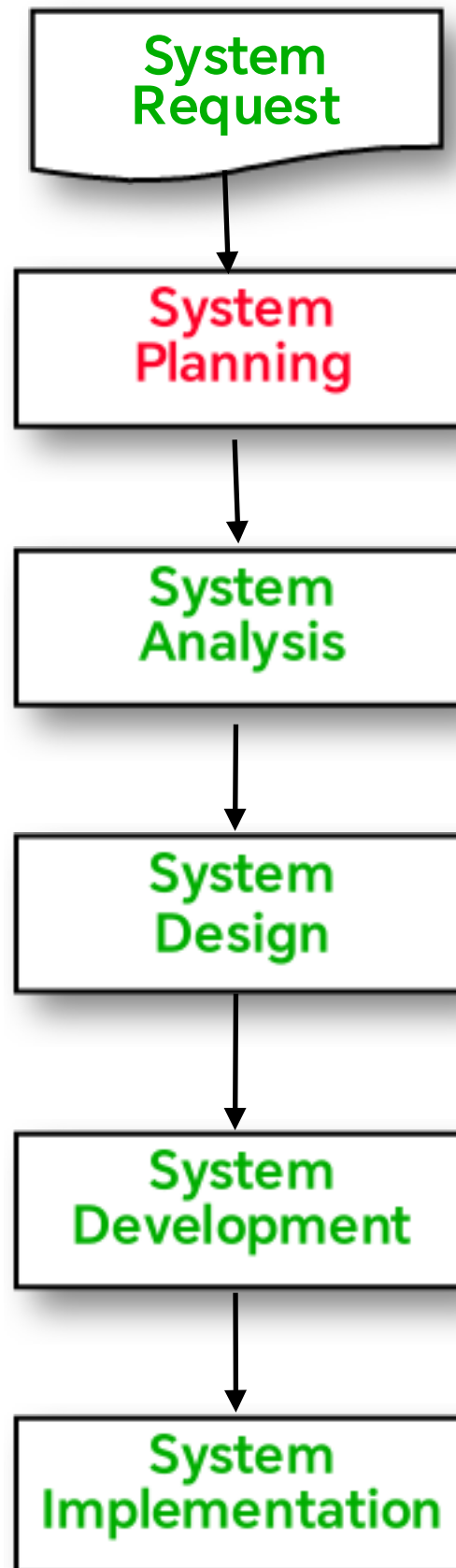


Chapter 2

System Planning

SDLC



First phase in
SDLC

System Request

- The SDLC starts from a 'System Request'. It may be a **formal** (a form) or informal (eg verbal instruction) request.
- **Purpose:** Request for **services** (eg develop a new system) from the IS department to solve problems.
- **Reasons:** for a System Request :
 - Existing system cannot support the new organisation's business **objectives**
 - Existing system has **problems**, errors or lacking in some functions, leading to not able to support some business processes.

Sample System Request

Tech Support Request System

Florida Institute of Technology

Submit Request

First Name: Last Name:

Telephone: Email-ID:

Describe the problem:

(Maximum of 4000 characters)

Date critical: ☐ Yes ☒ No

Submit Call Ticket

Manhattan Construction

Form IS-102A
REQUEST FOR SYSTEM SERVICES

SUBMITTED BY: Lynne Williams DATE: 10-15-01

DEPARTMENT: Marketing

TYPE OF REQUEST: ☐ New System
☒ Existing System Enhancement
☐ Existing System Modification

BRIEF STATEMENT OF PROBLEM (Attach additional documentation as necessary):

Customers are requesting we post daily progress reports on the Web.

BRIEF STATEMENT OF EXPECTED SOLUTION:

Our current Web site could be enhanced to include daily progress reports.

ACTION (To be completed by steering committee member):

☒ Request Approved Analyst Assigned: Rick Claremont
☐ Request Delayed Start Date: 10-29-01
☐ Request Reject Until: _____
Reason: _____

Signature: Andy Peterson Date: 10-23-01

Example of an online system request form

Sources of System Request

1. **From Top Management.** Instruction to improve on existing system to meet management' s requirements
 - New *reports* needed.
 - New organisational *objectives* eg. Increase sales, needs new modules (eg CRM system).
 - New *departmental* functions (eg customer service) needs new modules.
2. **From Users.** Users' needs for new features to support new roles or to solve existing system problems.
 - New features required for inward returns (inventory system).
 - New requirements to support new job roles in sales (eg, customer service staff)
3. **From Existing System Problems.** Errors or inefficiencies within existing systems as reported (by users) or observed (by IS personnels).
 - Poor system performance eg slow, **cannot** support more users

Sources of System Projects

4. From External Sources. Outside forces may influence the pressure for changes to existing systems.

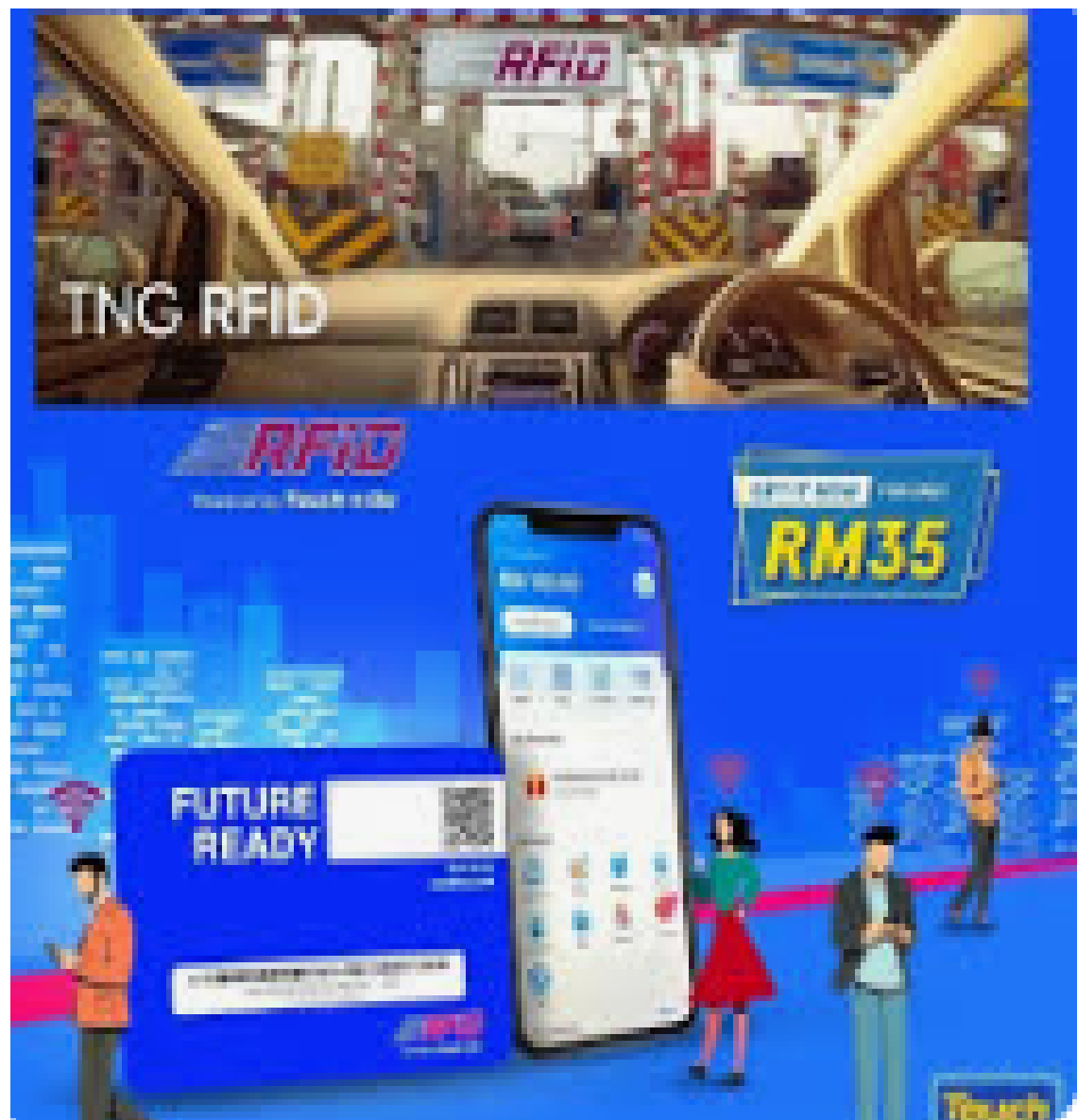
- **Technology.** New advances in technology put pressure to develop new systems eg. ERP system, RFID technology.
- **Regulations.** New government tax regulations requires changes be made to the payroll system
- **Customers.** New requirements by important customers who demands online sales features

5. From IS/IT Department. IS department's own recommendations for projects.

- IS personnel discovered poor performance of the existing system.
- IS Personnel found lots of errors and bugs.

payroll system generate payslip

Company (Malaysia) Sdn Bhd				Page 1 of 1			
548594 LIM				01 Jan 2013 - 31 Jan 2013			
				Basic Salary : MYR 8,166.67			
				Joined : 03 May 2010			
				Separation :			
				MYR	YTD	YTD(MYR)	
Earnings							
Basic Salary				8,166.67		8,166.67	
				-----		-----	
Total Earnings				8,166.67		8,166.67	
Deductions							
Employee EPF				(902.00)	(	902.00)	
Employee SOCSO				(14.75)	(	14.75)	
Income Tax				(814.20)	(	814.20)	
				-----		-----	
Total Deductions				(1,730.95)	(	1,730.95)	
Net Pay				6,435.72		6,435.72	
				=====		=====	
Employer EPF				984.00		984.00	
Employer SOCSO				51.65		51.65	
To Be Credited 22-806011 on 01 Feb 2013							
Annual	Entitlement	Earned	B/F	In-lieu	Taken	Balance	
	20.00	1.50	5.71	0.00	0.00	7.21	



<https://www.touchngo.com.my/>

- <https://www.linkedin.com/pulse/integration-rfid-library-management-software-r-sunil-kumar>



RFID Forklift Reader



RFID Fixed Reader



RFID Tag



RFID Handheld Reader



Middleware

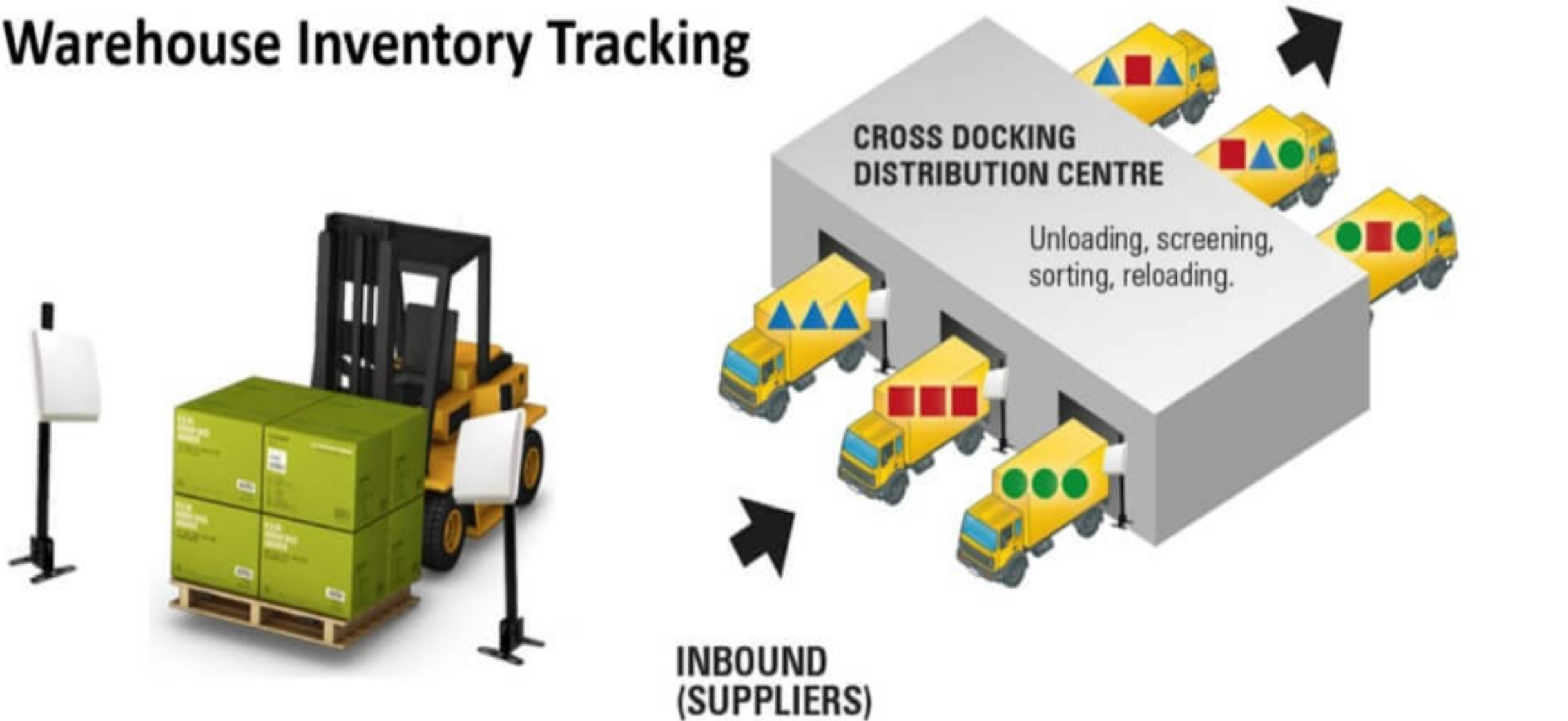


Quick Contactless RFID Inventory System, *Anti-Theft* Protection.
For any warehouses and assets.





Warehouse Inventory Tracking



Types of Improvement Required

3. **Improve Quality of Reports.** Information produced might be inadequate, incomplete, not timely or available.
 - Reports produced are **not** : *accurate, complete, relevant, useful, timely*
4. **Create Effective Controls.** Controls might be inadequate or excessive.
 - **Inadequate** – weak system access or log control leads to security problem
 - **Excessive** – data input control might be too excessive eg limit checks (upper or lower) slows down transaction.
5. **Reduce Cost.** Operation and maintenance costs in the current system are too high.
 - **Operation** – existing system and the department operation cost is high eg. cumbersome purchasing process (create PO) require lots of manpower.
 - **Maintenance** – existing system has lots of bugs, lack functions and features, lack documentation.

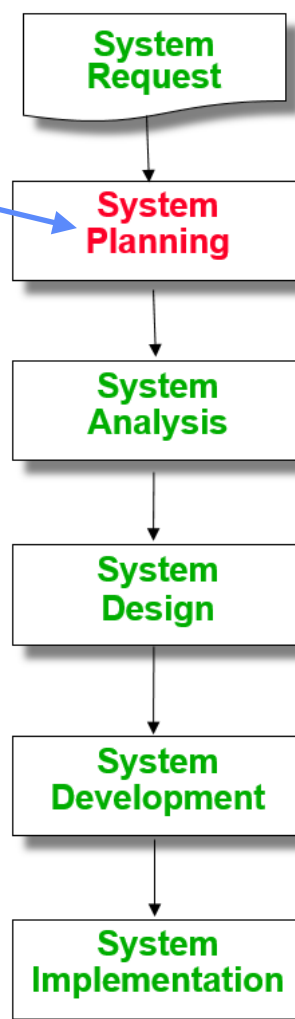
Types of Improvement Required

1. **Improve Performance** . The current system might not perform as well as it must.
 - **Reports** might take too long to prepare.
 - System cannot support increasing **number** of users
 - **Transactions** take a long time to complete.
2. **Improve Functions**. Improving functions of the system to both the internal end-users (staff) and external end-users (customers).
 - Simplifying overtime claims process for staff.
 - Improving share registration process for customers.
 - Add new functionalities to the system eg. create PO.
 - Providing customers with up-to-date product brochures.



Preliminary Investigation

- **When** : after a system project has been approved
- **Purpose** : gather information to determine if problems specified in the systems request *warrant* conducting subsequent phase of the SDLC ie System Analysis.
- **Areas Involved** :
 - Investigate about the system requested.
 - Determining the true nature and **scope** of the problem (system requested).
- **Output** : **Preliminary Investigation Report to management for approval**



Feasibility Study

Definition of Feasibility Study

- A study carried out :
 - To determine whether the new system is **viable /workable** or feasible (technically, socially and operationally, and economically).
 - **Example** : To determine whether the new system is :
 - economically **justifiable** eg. **benefits** outweigh the **cost** of developing and implementing the solution,
 - socially and operationally viable.

Types of Feasibility Study

1. Technical Feasibility

- Concern with
 - Specifying the **hardware**, **software**, and **human** requirements of the application system under study.
 - Evaluating whether the above requirements are **technically** achievable (ie. feasible).
- Issues to be addressed
 - **Number of users** – the ability to support increased number of users
 - **Data Inputs** – the ability to accurately enter and process a large number of inputs, such as examination marks
 - **Response time** – the ability to provide a given response time eg. respond to a request from a customer for an insurance quotation
 - **Outputs** – the ability to produce certain outputs, such as examination results, by a certain deadline.

2. Social and Operational Feasibility

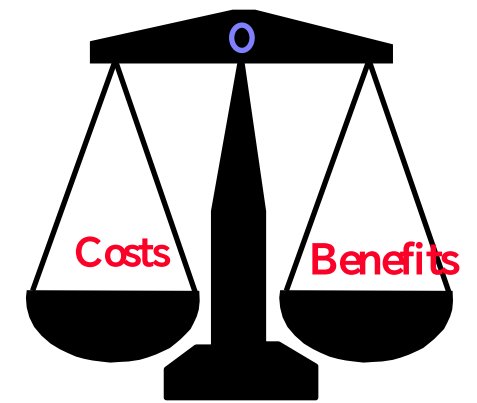
- **Concern with**

- Assessing whether the project can **fit** into the existing *social* and operational structure of the organization.
- Identifying any **additional costs** that may be incurred (eg cost of staff re-training, redundancies) or new benefits.

- **Issues to be addressed**

- **Employee skills** – the new system may require new skills and attitudes.
- **Employee motivation and resistance** – likely job changes caused by the system and how these might affect the motivation or resistance of employees.
- **Organisation Structural changes** – changes in management responsibilities, relationships, status and chains of command, or does not suit regional reporting structures.
- **High costs** – Assess direct (redundancies, retraining) and indirect (eg. in terms of morale and motivation of remaining staff) costs.

3. Economic Feasibility



- **Concern with**
 - Identify and quantify the various **costs** and **benefits** of implementing the proposed system.
 - **Evaluating** whether the project is economically viable/workable taking into accounts the costs and benefits identified.
- **Issues to be addressed**
 - **Technique.** Identify the various techniques to compare costs and benefits eg. ROI, IRR, NPV, payback etc.
 - **Selection.** Determine which of the technical options under consideration the company should choose.
 - **Decision.** Making decision whether to continue or halt/stop the project.

Project Costs (for system development)

Cost Groups	Cost Elements
Hardware	Server Clients (and other peripherals) Network Infrastructure
Software	Operating System Database Management System (DBMS) Programming tools
Development	System Planning System Analysis System Design System Development (Coding + Testing)
Implementation	Installation (for hardware and software) Data migration Training System changeover

Project Costs (for purchase/buy system)

Cost Groups	Cost Elements
Hardware	Server Clients (and other peripherals) Network Infrastructure
Software	Operating System Database Management System (DBMS) Programming tools Application software (eg Enterprise System)
Purchase	System Planning System Analysis (brief) Software Evaluation and Selection System Design (Business Process Requirement + configuration)
Implementation	Installation (for hardware and software) Data migration Training System changeover

Intangible Costs

- **Learning curve costs.** Staff need to learn how the system operates. During the learning curve, staff become slower and make more mistakes.
- **Dysfunctional costs.** Staff dissatisfaction and dysfunctional behaviour eg. scolding customers, come late.
- **Incompatibility costs.** Costs of incompatibility between existing systems and the new system. For example, replacing old system or build a special data file conversion facility.
- **Opportunity costs.** Whenever money is invested in one area, the opportunity to invest in another area is foregone.

Exercise for Students

Excel Manufacturing Ltd is planning to develop an enterprise system consisting of many modules eg sales, procurement/purchasing /buying , HR, Accounting, Production, Warehouse Management modules etc.

- a) List the 4 main **cost** elements for the *enterprise system*. For each element, provide some examples
- b) List 3 potential **benefits** of using the *procurement system*.

(a) Solutions – Costs

- **Software costs** – OS (eg Unix), DBMS (eg Oracle), Software development tools (eg visual studio, CASE tools)
- **Hardware costs** – Server, Clients (eg PCs), Network Infrastructure (eg. modem, cable)
- **Development costs** – System planning, system analysis, system design, system development.
- **Implementation costs** – Installation, data migration, training, changeover

(b) Solution – Potential Benefits

- Source for **alternative** suppliers and hence reduce costs of purchase
- Reduce **unplanned** (last-minute) purchases and hence save costs.
- Centralised purchasing in **bulks** reduce purchase costs through discounts.
- Proper **tracking** of purchases ensure timely delivery of purchased parts.
- Suppliers **rating** helps to select suppliers with better quality of purchased parts.
- Allow **just-in-time** (JIT) delivery which reduces stock holding cost.

Tangible or Quantifiable Benefits

- **Reduced operating costs.** Due to streamlining of processes, reduced office space.
- **Reduced staff costs.** Reduced number of staff required.
- **Reduced stock holding costs.** Through up-to-date information eg reduce stock holding costs.
- **Increased sales.** Through improved customer retention.
- **Increased productivity.** Reduced number of staff or increased output per staff.

Intangible or Non-Quantifiable Benefits_s

- Improved quality of **customer service** or satisfaction
- Improved company **image**
- Better information accessibility which leads to better **decision making**.
- More **accurate** management information.
- More timely and **up-to-date** information for decision makers.
- Improved staff **morale**, motivation and retention
- Greater reliability and **security**

Exercise for Students (Benefits)

- Implementing an *Inventory Control* system
 - List 4 tangible benefits
 - List 4 intangible benefits

Solutions (Benefits)

- **Tangible Benefits**
 - **Reduced staff costs.** Reduced number of staff required in warehouse
 - **Reduce Stock Holding Cost..** Reduced stock holding cost, losses from obsolescence (obsolete /old product) and stock-outs.
 - **Reduced operating costs.** Reduced warehouse space.
- **Intangible Benefits**
 - More **timely** and up-to-date information about inventory for warehouse and sales managers.
 - Improved staff **morale** and motivation of warehouse staff.
 - Faster **response time** of stock balances to Sales personnel.
 - Improved speed of **delivery** of goods to customers

Cost Benefit Summary

COST BENEFIT SUMMARY						
DATA FOR PLAN A: In-House Development						
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Costs:						
Consultant time to develop system	14,440					
Purchase necessary software platform	2,500					
Hardware	5,000					
Consultant training time	3,900					
Office staff maintenance, backup, and update	884	1,768	1,768	1,768	1,768	1,768
Total Costs:	26,724	1,768	1,768	1,768	1,768	1,768
Benefits:						
Eliminate overtime	995	1,989	1,989	1,989	1,989	1,989
Eliminate additional position	8,840	17,680	17,680	17,680	17,680	17,680
Eliminate daily errors	1,105	2,210	2,210	2,210	2,210	2,210
Total Benefits:	10,940	21,879	21,879	21,879	21,879	21,879

The above only intends to show an example of one of the **formats** for costs/benefits presentation. The items shown may **NOT** be suitable to your Assignment.

Cost Benefit Summary

$$17584 + 5268 = 22852$$

$$10940 + 21879 = 32819$$

DATA FOR PLAN B: Purchase Vertical Software Package						
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Costs:						
Purchase price for vendor package	3,000	3,000	3,000			
Tech support contract		500	500	500	500	500
Consultant time to install, configure, and test	4,800					
Hardware	5,000					
Consultant training time	3,900					
Office staff maintenance, backup, and update	884	1,768	1,768	1,768	1,768	1,768
Total Costs:	17,584	5,268	5,268	2,268	2,268	2,268
Benefits:						
Eliminate overtime	995	1,989	1,989	1,989	1,989	1,989
Eliminate additional position	8,840	17,680	17,680	17,680	17,680	17,680
Eliminate daily errors	1,105	2,210	2,210	2,210	2,210	2,210
Total Benefits:	10,940	21,879	21,879	21,879	21,879	21,879

The above only intends to show an **example** of one of the **formats** for costs/benefits presentation. The items shown may **NOT** be suitable to your Assignment.

Payback : profit from an investment equal to the initial cost.

Feasibility Report

1. *Aim and Objectives*

- should be clearly stated
- agreed upon by the managers
- areas involved are defined.

2. *Costs and Benefits Comparison*

Alternative solutions should be outlined indicating for each its respective cost/benefits comparisons using eg. Payback (profit from an investment equal to the initial cost.)

3. *Description of Proposed System*

Describe the hardware, software and business processes. Eg. DFD.

4. *Description of Current System*

Describe how the current system operate in the organisation. eg DFD.

5. *Development & Implementation Plan*

Includes a time-table and methods of change-over.

6. *Recommendations*

Final recommendation of a particular solution or methods.